

Chain and Independent Food Service Business Performance According to Income and Neighborhood in Boston



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Data Science Major Capstone

Background and Aims

- Food health and safety are essential so that illness does not spread.
- Investigating differences in food handling practices across neighborhoods helps better understand whether there are trends between socioeconomic class, neighborhood, and food safety.
- Previous study in Philadelphia and British Columbia, Canada show that chain restaurants have fewer total violations per inspection than non-chain restaurants (Tung, 2018). Non-chain restaurants generally have a higher proportion of people living in poverty, and residents live in lower income with more Black and Hispanic residents near them (Leinwand et al., 2017).
- This study is interested in the relationship between chain food inspection violations and the neighborhoods in which they reside.

Research Question

To what extent do chains perform differently (in terms of violations) than independent businesses in different neighborhoods based on income in Boston in 2024 to 2026?

Methods

- The original dataset consisted of one row per violation.
- Data were then spatially joined with census tract information. Coordinates of businesses were mapped to census tract polygons. Some businesses had null coordinates.
- Income data was then included and median income added to each row. There are 9 special purpose tracts (airports, schools, etc) and 4 tracts without income data.
- Data was then constrained to 2024-2026 and deduplicated so there is one row per inspection visit. Each visit has a separate column with the number of violations per visit.

Data Summary

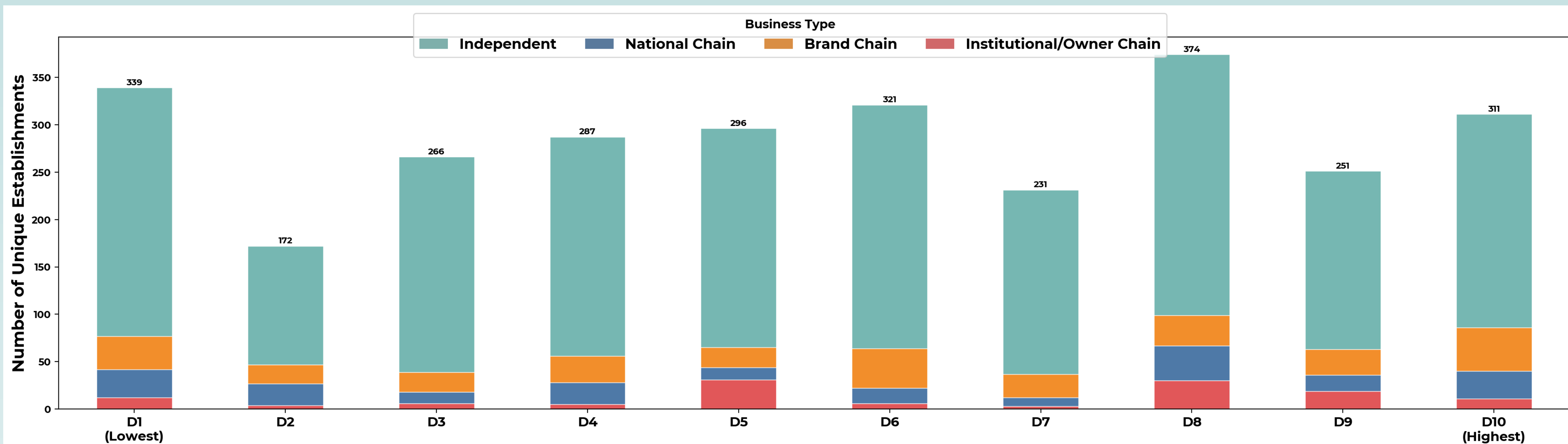


Figure 1: Unique Food Establishments by Business Type and Neighborhood Income Decile

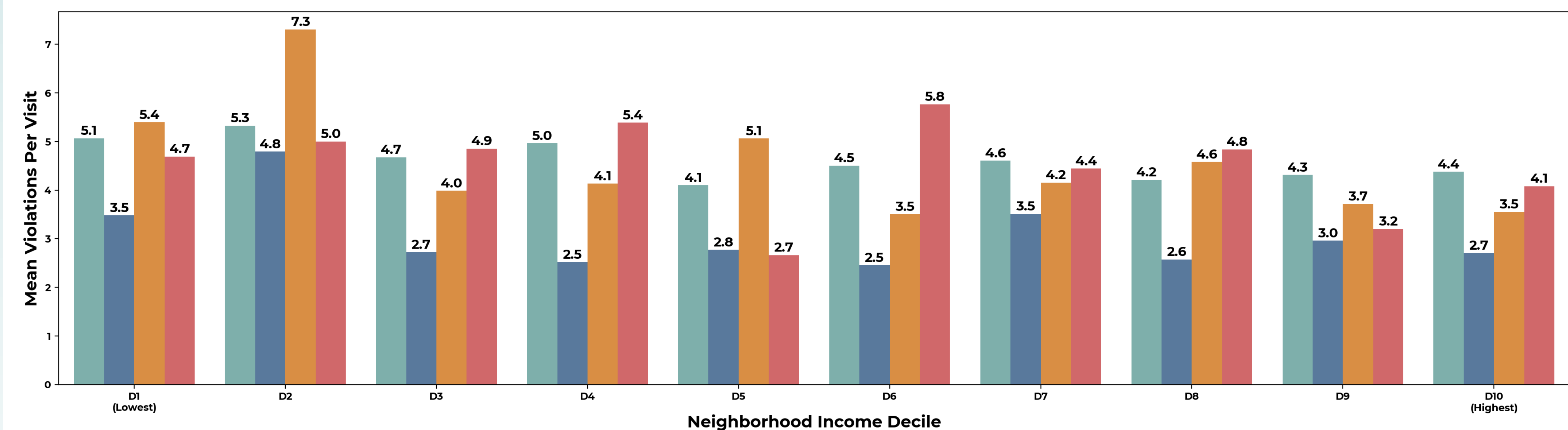


Figure 2: Mean Violations Per Visit by Business Type and Neighborhood Income Decile

Table 1: Negative Binomial Interaction Model Results

Variable	Coefficient	Incidence Rate Ratio	P-Value
Intercept (baseline)	1.405586	4.08 violations per visit, baseline	< 2e-16
National Chain	-0.392238	32.5% fewer	1.95e-07
Brand Chain	0.206467	22.9% more	0.00907
Institutional/Owner Chain	-0.104277	9.9% fewer	0.38596
Median Income	-0.006885	0.7% decrease with \$10k	0.07231
National Chain x Median Income	0.001685	0.17% more with \$10k increase	0.77128
Brand Chain x Median Income	-0.017272	1.7% decrease with \$10k increase	0.00673
Institutional/Owner Chain x Median Income	0.004031	0.4% increase with \$10k increase	0.66080
Conditional R ²	0.399	Marginal R ²	0.028

Results and Discussion

- There are fewer violations in high income neighborhoods than there are in low-income neighborhoods (Figure 2, Table 1).
- Negative binomial mixed effects model was used due to overdispersion in the Poisson.
- Random effects include census tract and establishment. Fixed effects are analysis group and median income.
- Significant findings are as follows: National chains have 32.5% fewer violations than independents when income is \$0.
- Brand chains have 22.9% more violations than independents when income is \$0.
- For every \$10,000 increase in median household income, brand chain violations decrease by 1.7%.
- Independents have a predicted 4.08 violations compared to average of 4.45 violations.
- Fixed effects explain 2.8% of the variance (Marginal R²) and the full model explains 39.9% of the variance (conditional R²).

Limitations

- There is potentially temporal and spatial correlation in the model. Two tracts next to each other probably have more similarities than another on the other side of the city. The model does not account for trends over time.
- Independents comprise the majority of inspection visits. As such, it is much more difficult to make accurate predictions for chains (Figure 1).
- A few chains dominate the National Chain category, meaning that these few chains' outcomes represent the entire group.

References

- Food Establishment Inspections. (n.d.). [Tabular]. Inspectional Services Department. Retrieved <https://data.boston.gov/dataset/food-establishment-inspections>
- Leinwand, S. E., Glanz, K., Keenan, B. T., & Branas, C. C. (2017). Inspection Frequency, Sociodemographic Factors, and Food Safety Violations in Chain and Nonchain Restaurants, Philadelphia, Pennsylvania, 2013-2014. *Public Health Reports*, 132(2), 180-187. <https://doi.org/10.1177/0033354916687741>
- TIGER/LineShapefiles.(2023).[Dataset].<https://www.census.gov/cgi-bin/geo/shapefiles/index.php>
- Top 500 Chains. (n.d.). Restaurant Business. Retrieved <https://www.restaurantbusinessonline.com/top-500-2020>
- Tung, J., BCIT School of Health Sciences, Environmental Health, & Heacock, H. (2018). Critical food safety violations in Surrey: relationship to community median household income and restaurant type. *BCIT Environmental Public Health Journal*. <https://doi.org/10.47339/ephj.2018.57>
- U.S. Census Bureau, American Community Survey. (2024). *S1901: Income in the Past 12 Months (in 2024 Inflation-Adjusted Dollars)* [Dataset]. <https://data.census.gov/all?q=S1901>